4.2 Space, time and motion

This section develops classical mechanics, including vectors. The conservation of momentum, the kinematics of uniformly accelerated motion and the dynamics of motion in two dimensions under a constant force are covered. IT skills may be developed through a variety of data capture techniques and simple mathematical modelling (HSW3).

	Learning outcomes	Additional guidance
(a)	*Describe and explain:*	
	(i) the use of vectors to represent displacement, velocity and acceleration	
	(ii) the trajectory of a body moving under constant acceleration, in one or two dimensions	
	(iii) the independent effect of perpendicular components of a force	
	(iv) calculation of work done, including cases where the force is not parallel to the displacement	
	(v) the principle of conservation of energy	HSW1
	(vi) power as rate of transfer of energy	
	(vii) measurement of displacement, velocity and acceleration	
	(viii) Newton's laws of motion	HSW7
	(ix) The principle of conservation of momentum; Newton's Third Law as a consequence.	HSW2, 7
(b)	*Make appropriate use of:*	
	(i) the terms: displacement, speed, velocity, acceleration, force, mass, vector, scalar, work, energy, power, momentum, impulse	
	by sketching and interpreting:	
	(ii) graphs of accelerated motion; slope of displacement–time and velocity–time graphs; area underneath the line of a velocity–time graph	*M3.3, M3.4, M3.5, M3.6, M3.7, M3.12* HSW5
	(iii) graphical representation of addition of vectors and changes in vector magnitude and direction.	

(c) *Make calculations and estimates involving:*

(i)	the resolution of a vector into two components at right angles to each other	*M4.5*
(ii)	the addition of two vectors, graphically and algebraically	algebraic calculations restricted to two perpendicular vectors *M4.1, M4.2, M4.4*
(iii)	the kinematic equations for constant acceleration derivable from: $a = \frac{v-u}{t}$ and average velocity $= \frac{v+u}{2}$: $v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$	*M2.2, M2.4*
(iv)	momentum $p = mv$	*M2.3*
(v)	the equation $F = ma = \frac{\Delta(mv)}{\Delta t}$ where the mass is constant	Learners will also be expected to recall the equation *F = ma* *M2.1*
(vi)	the principle of conservation of momentum	one-dimensional problems only
(vii)	work done $\Delta E = F\Delta s$	If displacement is at an angle θ to the force $\Delta E = F\Delta \cos\theta$ *M2.1*
(viii)	kinetic energy$= \frac{1}{2}mv^2$	Learners will also be expected to recall this equation *M0.5*
(ix)	gravitational potential energy = *mgh*	Learners will also be expected to recall this equation *M2.3*
(x)	force, energy and power: power $= \frac{\Delta E}{t}$, power = *Fv*	*M2.3*
(xi)	modelling changes of displacement and velocity in small discrete time steps, using a computational model or graphical representation of displacement and velocity vectors.	calculations restricted to zero or constant resultant force *M3.9*

(d) *Demonstrate and apply knowledge and understanding of the following practical activities (HSW4):*

(i)	investigating the motion and collisions of objects using trolleys, air-track gliders etc. with data obtained from ticker timers, light gates, data-loggers and video techniques	links to **4.2a(vii), b(ii), c** **PAG1**
(ii)	determining the acceleration of free fall, using trapdoor and electromagnet arrangement, lightgates or video technique	links to **4.2a(vii), b(ii), c** **PAG1**
(iii)	investigating terminal velocity with experiments such as dropping a ball-bearing in a viscous liquid or dropping paper cones in air.	links to **4.2a(vii), b(ii), c** **PAG1**